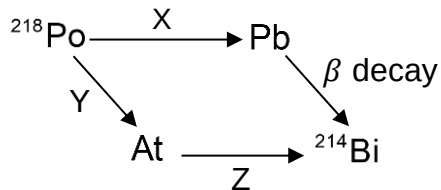


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^{218}Po decays to ^{214}Bi via two pathways as shown in the figure below.



What are the possible radioactive decay modes X, Y and Z?

		X	Y	Z
	A	γ decay	α decay	β decay
	B	α decay	β decay	α decay
	C	β decay	α decay	γ decay
	D	α decay	α decay	β decay

L2 Answer: B

Decay mode X must be an α decay because Pb undergoes a β decay to ^{214}Bi which only changes the atomic number and not the mass number.

[Turn over

To decay from ^{218}Po to Pb to ^{214}Bi there must be a decrease in the mass number by 4, from 218 to 214. α particle is ${}^4_2\text{He}$ which can account for this decrease in mass number.

Options A and C are out.

Option D is out because if decay mode Y is an α decay, the daughter nuclide would be Pb and not At.

END OF PAPER